

Computational Realization of Lambda-Field Coherence Dynamics

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Abstract

Supplies the executable layer between Lambda-field theory (100.007) and local open-system reciprocity (100.009). This v2.0 release keeps Lambda as an independent dynamical field — state, coherence target, and field are separate objects, the minimal anti-circularity control — and adds fixed-seed reference implementations: a finite-difference Lambda engine and a phase-reference oscillator network. The pilots make the 100.007 -> 100.008 -> 100.009 bridge executable and falsifiable, with explicit kill conditions, without claiming a proof of physical ontology.

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1. Role in the 100-series

[[FRC-100-007]] introduces the Lambda-field as a scalar completion of coherence dynamics. [[FRC-100-009]] writes local open-system reciprocity as a transport law. This paper is the computational bridge: 100.007 Lambda field -> 100.008 executable Lambda dynamics -> 100.009 local reciprocity.

The role of 100.008 is not to introduce another ontology. Its job is to say how the Lambda-field can be represented, updated, diagnosed, and connected to oscillator implementations **without collapsing the field into an algebraic restatement of the state**. The v1 paper identified the main danger: if Lambda is defined pointwise as a direct function of a local entropy density, it has no independent relaxation, no memory, and no legitimate field dynamics. The v2 paper keeps that correction and adds a reproducible reference implementation.

2. State, Target, and Field Are Separate Objects

Let the computational state be a normalized density $\rho(x,t) \geq 0$, $\int_{\Omega} \rho dx = 1$. The theory uses four separate objects: the state field ρ ; the mesoscopic coherence target $C_K[\rho]$; the equilibrium target field $\Lambda_{\text{eq}}[\rho]$; and the actual dynamical field Λ .

The separation matters. If one writes, pointwise,

$$[C(x,t) = \exp[-s(x,t)/k_*], \quad \Lambda(x,t) = \Lambda_0 \ln C(x,t),]$$

then Λ becomes only $-\Lambda_0 s/k_*$ — a mirror of the state, not a dynamical field. A mirrored variable cannot honestly carry its own diffusion constant or relaxation time. This is the anti-circularity requirement of the whole paper.

Here k_* is a scale-setting reciprocity parameter — a units bridge, calibrated in the pilot from the mean kernel score of the run, so it is scale-normalized rather than treated as a fitted universal constant.

3. Kernel Coherence Target

The local target is mesoscopic, not pointwise. With a smoothed score

$$\bar{s}_K(x,t) = \int_{\Omega} K_{\ell}(x-y) s_{\gamma}(y,t) dy, \quad s_{\gamma} = -\rho \ln(\rho + \epsilon) + \frac{\gamma}{2} |\nabla \rho|^2,$$

the kernel coherence target and equilibrium Λ target are

$$C_K[\rho] = \exp[-\bar{s}_K/k_*], \quad \Lambda_{\mathrm{eq}}[\rho] = \Lambda_0 \ln C_K[\rho].$$

Following [FRC-566-030], the closed identity $dS + k_* d \ln C = 0$ must not be used as the anchor for open dynamics; the operational object is the open transport loop.

4. Dynamic Λ Equation

The actual Λ field obeys a relaxation-diffusion equation, and the state responds to the field:

$$\partial_t \Lambda = D_{\Lambda} \nabla^2 \Lambda - \mu_{\Lambda}(\Lambda - \Lambda_{\mathrm{eq}}[\rho]) + J_{\mathrm{ext}},$$

$$\partial_t \rho = D_{\rho} \nabla^2 \rho - \alpha \nabla \cdot (\rho \nabla \Lambda) + \Xi_{\rho}.$$

This produces the computational loop $\rho \rightarrow C_K[\rho] \rightarrow \Lambda_{\mathrm{eq}}[\rho] \rightarrow \Lambda \rightarrow \rho$: reciprocally coupled but not algebraically identical.

5. Reference Implementation A: Finite-Difference Λ Engine

On a periodic 1-D grid, each step computes the local score, smooths it by FFT convolution with a Gaussian kernel, calibrates k_* , computes C_K and Λ_{eq} , updates Λ spectrally, updates ρ by diffusion plus Λ -guided drift, then clips and renormalizes. The pilot uses two Λ source windows as attractor regions. Measured fixed-seed metrics:

Diagnostic	Initial	Final	Change
Entropy S	5.544985	5.534553	-0.010432
Kernel coherence $\langle C_K \rangle$	0.367922	0.369636	+0.001714
Λ -target mismatch E_{Λ}	1.001261	0.156237	-84.4%
Attractor-window mass	0.295593	0.355097	+0.059504
Concentration $\max(\rho)/\mathrm{mean}(\rho)$	1.036124	1.461361	+0.425237

The strongest result is not the small entropy/coherence shift. It is the explicit field separation: the target is computed from ρ , the actual Λ field relaxes toward it with lag, and the state then responds to the field gradient. This is the minimal anti-circularity control.

6. Oscillator Implementation

Λ can also be realized by oscillator dynamics on phases ϕ_i with Kuramoto base coupling. A spatial vector $\nabla \Lambda$ cannot be added directly to a scalar phase rate; three scalar channels are admissible:

- **Channel A — scalar frequency bias:** add $\eta \Lambda(x_i, t)$ to the phase rate.
- **Channel B — phase-reference coupling (preferred):** with reference $L(x, t) = C \Lambda e^{i\theta_\Lambda}$,

$$\dot{\phi}_i = \omega_i + \sum_j J_{ij} \sin(\phi_j - \phi_i) + \eta C \Lambda(x_i, t) \sin(\theta_\Lambda(x_i, t) - \phi_i).$$

- **Channel C — projected gradient drive:** $\eta \mathbf{n}_i \cdot \nabla \Lambda(x_i, t)$, only when a physical orientation vector $\mathbf{n}_i$ exists.

The phase-reference channel makes Λ a coherence clock rather than a dimensionally invalid vector injection.

7. Reference Implementation B: Phase-Reference Locking

A ring network with local Kuramoto coupling and a fixed Λ phase-reference field checks the corrected scalar channel:

Diagnostic	Initial	Final	Change
Global order R	0.172888	0.912092	5.28x
Reference error	1.059117	0.053094	-95.0%
Phase spread	1.781577	0.424640	-1.356936

This test is intentionally modest. It proves the preferred oscillator channel is executable and dimensionally consistent. It does not prove that physical measurement is oscillator locking.

8. Relation to 566.030 and 100.009

The original 100.008 pointed to 566.001; that anchor is stale. The correct anchor is the open 566.030 frame: in open systems the combined ledger is not a closed conservation identity, and the residual

must be measured through export channels. 100.008 should be read as an **engine, not a thermodynamic proof**. It supplies fields and fluxes that [[FRC-100-009]] can place into a local open-system law

$$\partial_t s + k_* \partial_t \ln C + \nabla \cdot J_R = \Sigma_R .]$$

The open reciprocity papers decide which candidate transport objects (ρ , Λ , C_K , field mismatch, drift flux, convergence time, oscillator order) survive as measured physical channels.

9. Kill Conditions

This paper fails in its computational role if:

1. Λ is defined as an instantaneous pointwise algebraic function of ρ ;
2. Λ cannot be updated independently of $\Lambda_{\rm eq}$;
3. the finite-difference engine fails positivity or normalization;
4. E_Λ does not decrease in the relaxation pilot;
5. attractor-window mass does not increase in the drift pilot;
6. oscillator coupling adds a spatial vector directly to a scalar phase rate;
7. the phase-reference pilot does not increase order or reduce reference error.

The shipped pilots pass these checks, but passing them is not enough to claim physical ontology.

10. Appendix Program and Source Package

Next steps: a 2-D Λ engine with attractor basin maps; parameter sweeps for ℓ , μ_Λ , D_Λ , α , and k_* calibration; an explicit open export channel compatible with [[FRC-100-005]] v2; coupling to the [[FRC-100-003]] stage-1/stage-2 collapse structure; and a comparison between oscillator locking time and the [[FRC-787-787]] coherence-lifetime criterion.

The release package (`pilot/pilot_lambda_engine.py`, `pilot/lambda_metrics.json`, `pilot/compare_metrics.py`, figures, checksums) generates all quoted numbers from fixed seeds.

Connections

- [[FRC-100-007]] — The Λ -field this paper makes executable
- [[FRC-100-009]] — Local open-system reciprocity, the downstream transport law
- [[FRC-100-003]] — Collapse as open-system phase-locking; target of future engine coupling
- [[FRC-100-005]] — Thermodynamics of locking; open export channel compatibility

- [[FRC-566-030]] — The open-ledger frame replacing the stale closed-identity anchor
- [[FRC-787-787]] — Lambda-Flight; the oscillator locking time meets its coherence-lifetime criterion

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